MAE3405 Aerospace Propulsion

Dr Daniel Duke
Department of Mechanical & Aerospace Engineering
Monash University, Australia

Topic 5-6 Problem Set Questions

An asterisk indicates that this problem is similar to a past exam question.

Question 12

An engine with a mass flow rate of 0.1 kg/s has an inlet manifold pressure boost of 1.3 provided by a supercharger. If flying at an altitude of 2000 m;

(i) determine the power required to drive the supercharger assuming a mechanical efficiency of 90% and isentropic efficiency of 80%.

(ii) If an intercooler was used to reduce the supercharger temperature increase by 50%, determine the percentage increase in charge air density relative to atmospheric.

[Ans: (i) 3.0 kW; (ii) 24%]

Question 13

An in-line 6-cylinder water cooled, 4-stroke aircraft engine of 75mm bore and 100mm stroke has a brake power output of 110kW at 2800RPM. The volumetric efficiency at this operating condition referred to SSL is 80%. The engine is now fitted with a mechanically driven supercharger with an isentropic efficiency of 70% and pressure ratio of 1.6. The supercharged version has a volumetric efficiency of 100% referred to supercharger delivery pressure and temperature. Assume the indicated power developed per unit volume flow rate of induced air at ambient conditions is the same for normal aspiration and supercharging, and that the addition of the fuel can be ignored. Calculate the net increase in brake power resulting from the supercharger. Take the mechanical efficiency of the engine at 80% in both cases and the mechanical efficiency of the supercharger drive as 95%.

[Ans: 70.4 kW]

Question 14*

A 2m diameter propeller is being tested on the 4 cylinder SI engine whose data is listed below runs on avgas. If the test is done at an altitude of 2000m, determine the static thrust developed. Assume the engine operates on the fuel/air Otto cycle.
[Ans: 3.07 kN]

Piston Engine			
Quantity	**Value**	**Quantity**	**Value**
Compression ratio	7.5	Combustion efficiency	99%
Bore	0.1m	Mechanical Efficiency	85%
Stroke	0.125m	Volumetric efficiency	80%
RPM	2700	Air to fuel ratio	14.5
Cp reactants	1.055kJ/kg-K	Cp products	1.486kJ/kg-K
Cv reactants	0.768kJ/kg-K	Cv products	1.199kJ/kg-K

Question 15*

A helicopter hovers at 1000m. It has a mass of 750kg. Its main rotor is made up of two blades each having a length of 4m. The tail rotor also has two blades each having a length of 0.75m. If the power required to drive the tail is 10% of that required to drive the main rotor and the hovering condition can be considered a static thrust case, determine:

(i) the volumetric fuel consumption per hour,

(ii) the brake specific fuel consumption in kg/kW.h and

(iii) the thermal efficiency of the engine.

Specifications of the SI 4-stroke engine are listed in the table below.

Quantity	**Value**	**Quantity**	**Value**
Bore	0.1m	Combustion efficiency	98%
Stroke	0.12m	Mechanical Efficiency	90%
RPM	2500	Volumetric efficiency	85%
Cylinders	6	Air to fuel ratio	Stoichiometric
Fuel approximated by	$CH_{2.2}$	Fuel LHV	Avgas
Cp (fuel & air)	1.055kJ/kg.K	Cp (exhaust products)	1.486kJ/kg.K
Cv (fuel & air)	0.768kJ/kg.K	Cv (exhaust products)	1.199kJ/kg.K

[Ans: (i) 37.8 L/hr, (ii) 0.41 kg/kW.h, (iii) 22.9%]

Question 16D

A quadcopter drone hovers at an altitude of 200 ft. The drone maintains a steady position in a constant horizontal wind of 36 km/hr. The drone has a rotor diameter of 4 inches, a gross mass of 1.5 kg, and a drag coefficient of 0.75. The brushless DC electric motors have a KV rating of 1200 and are operating at 1/6 of their maximum torque with an efficiency of 80%. Given a total battery capacity of 1.3 Amp-hour at 22.2 Volts, calculate:

(i) how long the drone can maintain position, and
(ii) the rotor RPM.

[Ans: (i) 7.4 minutes, (ii) 22,200 RPM]